

GAN-SMOTE: A Generative Adversarial Network approach to Synthetic Minority Oversampling

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Abstract:

Small unbalanced datasets remain a serious impediment to the implementation of cutting-edge artificial intelligence in an industry setting. This paper proposes GAN-SMOTE, a novel approach to synthetic minority class oversampling using a generative adversarial network that can be applied to boost the performance of classifiers learning from small and unbalanced datasets. This paper also introduces techniques that are key for ensuring the stability and variance of the generative adversarial network. This method demonstrates meaningful improvement on a well-studied petrographical dataset with significant class imbalance.

Keywords: GAN-SMOTE, synthetic data, minority class oversampling, generative adversarial network, unbalanced dataset, small dataset, minibatch discrimination

1 Introduction

In recent years, machine learning techniques have been applied to problems in many different domain areas. As computer scientists begin to encounter problems in areas they are unfamiliar with, they will need to gain the skills to work with data that is beyond the realm of their own expertise. Organisations like Google AI and start-ups like Lobe are aiming to provide Artificial Intelligence as tools directly to domain experts so that the involvement of computer scientists is only required for complex or difficult problems. Providing solutions like these will require good ‘rules of thumb’ when dealing with small and imbalanced real-world datasets and may be the rate limiting step in the adoption of ‘AI as a Service’ products by domain experts.

While small datasets present a large problem for machine learning, due to effects such as sampling errors and issues with generalisation, these problems are exacerbated when class imbalance is present. Class imbalance occurs when there is an uneven number of samples for one or more of the target categories in a dataset and can skew the output of the model towards the larger class in order to maximise overall accuracy and minimise loss.

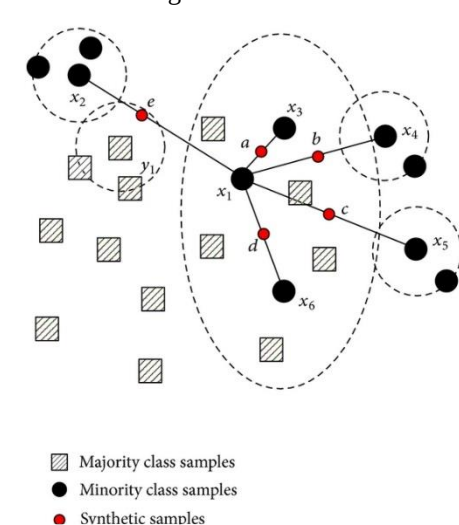


Figure 1 – SMOTE creates synthetic datapoints utilising existing relationships between minority class samples (Hu & Li, 2013).

There are several traditional techniques for handling class imbalance. The first is under sampling, which involves removing instances of the majority class so there is an even number of samples in each class. Although this fixes the imbalance problem the classifier misses out on valuable data about the majority class, and also risks further sampling error in small datasets. Another method is oversampling, where instances of the minority class are duplicated so that the classes become evenly distributed. This solves the balance issue but can cause the classifier to generalise worse for the minority class because the idiosyncrasies of the minority class data become more common. In neural networks, data scientists can manipulate the loss function to penalise misclassification of the minority class, although in effect this has the same issues as oversampling.

Another interesting technique is SMOTE (Synthetic Minority Oversampling Technique) (Chawla et al., 2002). SMOTE is a statistical technique that over-samples the minority class in imbalanced datasets by creating synthetic minority class examples. This technique boosts class membership whilst introducing enough variety to encourage classifier generalisation. Whilst these samples may not exactly reflect the real data distribution, they tend to be close enough to encourage useful amounts

of generalisation and boost overall classification accuracy. See Figure 1 for a diagram explaining SMOTE.

SMOTE has demonstrated great results in unbalanced datasets, but relatively limited in application due to its statistical nature. For example, without understanding the deeper meaning in highly structured data, SMOTE can't hope to create synthetic text samples for natural language processing tasks or realistic images for machine vision training.

This paper extends Scott's previous paper 'Classifying poorly understood datasets: the effect of encoding density and quality in shallow neural network classification' (Scott, 2019) and aims to improve the performance of the simple neural network on the paper's small imbalanced dataset using a novel technique.

This paper proposes GAN-SMOTE, an updated approach to synthetic oversampling using a generative adversarial network.

This novel technique is tested on a dataset of petrographical descriptions of core samples from the North West Shelf in offshore Australia (Gedeon et al., 2001). The class for prediction in this dataset is the sample's porosity, rated as either Very Poor, Poor, Fair or Good. The factors provided to determine the sample's porosity are detailed lithographical descriptions summarised into six characters: grain size, sorting, matrix, roundness, bioturbation and laminae. The data was provided in a sparse 'one-hot' encoded format, with a 1 in the relevant column when a sample has the associated attribute for a character. Scott's paper found that the one-hot encoding for the data yielded the best results, and so this original encoding is retained for use in this study.

The dataset is very small, consisting of 140 samples in total. It is split into a training and testing set of 70 samples each. The distribution of the attributes is also very uneven which can make classification difficult for outlier samples. Figure 2 details the distribution of attributes.

This paper will outline the architecture of GAN-SMOTE, including the techniques used to create a stable GAN for data synthesis.

This paper will then compare training-set GAN-SMOTE supplementation and pure GAN-SMOTE results to unaltered benchmark training set results.

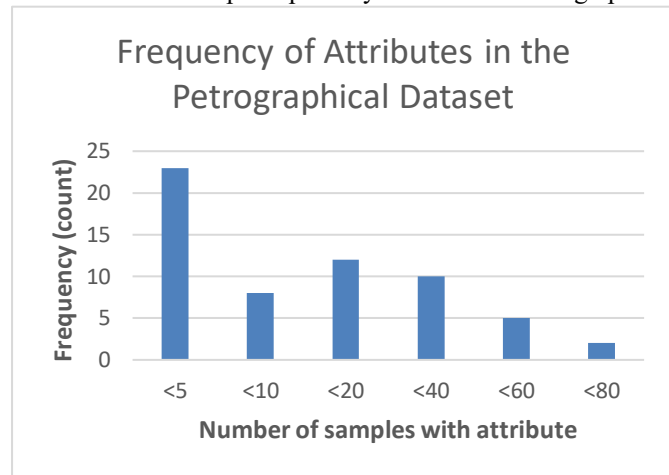


Figure 2 – Frequency distribution of features in the petrographical dataset, demonstrating “sparse” nature of the data.

2 Method

2.1 GAN-SMOTE Basic Architecture

The GAN-SMOTE architecture is based upon the original GAN architecture proposed in Goodfellow's 2014 paper 'Generative Adversarial Networks' (Goodfellow et al., 2014). The GAN-SMOTE architecture was in part inspired by the paper "Deep learning approach to generate offline handwritten signatures based on online samples" where an autoencoder was used to synthesise offline signatures to increase the dataset for signature fraud detection (Melo et al., 2019).

The GAN-SMOTE is made up of two different neural networks, a generator and a discriminator. The generator is a neural network with two hidden layers that takes 16 neurons of random input and produces 59 outputs. These outputs correspond to the 58 features in the dataset, plus an extra neuron to enable mini-batch discrimination. This implementation of mini-batch discrimination will be described in '2.3 Improving GAN-SMOTE Variance'.

The discriminator takes 59 neurons of input, corresponding to the 59 neurons of output from the generator. It has two hidden layers, and one output neuron.

In every epoch, the generator produces synthetic datapoints. The discriminator is then shown a set of real datapoints from a specific class of the data and predict if the data points were made by the generator. The correct output in this first part should be a score of 0.0 for each data point. The results are used to calculate loss, which is backpropagated through the discriminator. The discriminator is then shown a set of synthetic datapoints created by the generator. In this round the correct output should 1.0 for each data point. The results are again used to calculate loss, which is backpropagated through the discriminator. The generator is trained in a similar manner. The generator produces synthetic data, which is used as input to the discriminator. The results are used to calculate loss, which would be the inverse of the loss for the discriminator. In effect, the two networks are engaged in a 'minimax game', where each is rewarded for the failure of the other. Over many epochs of training, the discriminator improves its ability to recognise the generator's samples, and the generator improves its ability to produce realistic synthetic data samples.

At the end of training, the generator can be used to produce an arbitrary number of synthetic data samples for a particular class. In the case of the one-hot encoded petrographical dataset, this data was rounded up to 1 or down to 0 for consistency.

2.2 Improving GAN-SMOTE Stability

One of the most difficult tasks when training a GAN is ensuring that the network remains stable. When creating the GAN-SMOTE architecture, a number of changes had to be made to ensure that the networks were competitive with one another. The discriminator often out-performed the generator which caused the generator to stop improving. This is caused by an unchanging loss function which reflects the discriminator's 100% accuracy.

The first change that was made was the introduction of random noise to the input data. As a result of the one-hot encoding the input data is encoded as either 1.0 or 0.0, so any uncertainty in the generator made it obvious that the sample was synthetic. Because a rounding activation function has a gradient of 0, the output of the generator couldn't be coerced to produce only rounded numbers. As an alternative, the real data was altered with a random amount of noise between 0% and 2%.

Another technique that helped the network remain stable is random bit-flips. As each sample was an array of binary numbers, random noise alone won't prevent the discriminator from overfitting on the real data (the discriminator quickly learns that 0.97 is the same as 1.0). A predetermined proportion of the real data had the binary data reversed so that 1's became 0s and vice versa. This helps the generator remain competitive with the discriminator at the cost of the synthetic data's accuracy. The amount of noise is then slowly reduced to zero over the course of many epochs, allowing the generator to produce more accurate samples by the end of training.

The learning rate was adjusted in line with "SGDR: Stochastic Gradient Descent with Warm Restarts" (Loshchilov & Hutter, 2017) to prevent the networks from settling into local minima. The learning rate was increased by a factor of 10 and degraded back to the original learning rate using cosine annealing over the course of 1000 epochs. This is repeated for every 1000 epoch interval. This was particularly useful for ensuring the generator consistently improved. See figure 3 for visualisation.

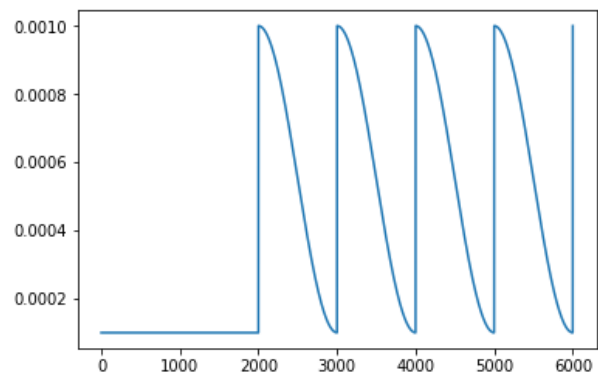


Figure 3 – Learning rate changes in 1000 epoch intervals, demonstrating the cosine annealing technique for warm restarts.

2.3 Improving GAN-SMOTE Variance

The GAN-SMOTE architecture had issues with 'mode collapse', where the generator ignores input and produces the same output each time. Minibatch discrimination was implemented as described in "Improved Techniques for Training GANs" [10]. The Manhattan distance was computed between each sample in a mini-batch and every other sample, and then the sum of these distances is appended to the output of the generator. This additional information about the distribution of the samples in the batch allows the discriminator to identify and penalise mode collapse, and encourages the generator to achieve a similar distribution of data to the real samples. Because of the small amount of data, it was necessary to add a certain amount of random noise (30%) to each individual sample so that the generator remained competitive with the discriminator.

Optimising the GAN-SMOTE architecture requires consideration of many different metrics. Balancing the GAN was possible due to a large amount of live data that allows the user to observe and tweak the operation of the model rapidly. Network losses and accuracy, visualisation of the data produced by the network using PCA, and samples of the synthetic and real data allowed rapid updates to hyperparameters affecting training stability, while sum of Manhattan distances within synthetic and real data batches, and sum of Manhattan distances between the datasets allowed fine-tuning of the synthetic data's variance and similarity to the real data.

2.4 Benchmarking GAN-SMOTE performance

A simple neural network, as described in 'Classifying poorly understood datasets: the effect of encoding density and quality in shallow neural network classification' [5] was implemented. The simple neural network had one hidden layer of 100 neurons, a cross entropy loss function and a stochastic gradient descent optimiser.

The original 70/70 train-test split of the petrographical data from Scott’s ‘Classifying poorly understood datasets: the effect of encoding density and quality in shallow neural network classification’ was used, and the training set was used to produce synthetic data for each of the 4 classes using GAN-SMOTE.

In the first experiment, the synthetic data was used to “top up” class imbalances in training data. In the next experiment, 100 synthetic class samples for each class was used to train the simple neural network.

Benchmark tests were performed using unaltered training data and a basic oversampling technique performing random sample duplications of minority classes to top up class imbalance.

The training data was somewhat imbalanced. The classes had 18, 22, 11 and 19 samples respectively. In the first synthetic “top up” experiment and basic oversampling experiment, each class was topped up to 22.

3 Results and Discussion

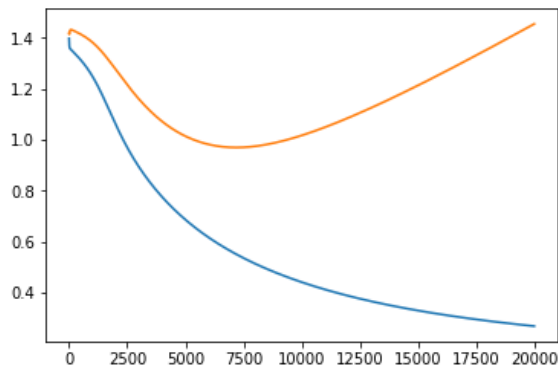
The results of the benchmarking tests are shown in Figure 4 and Figure 5.

	Synthetic “top up”	Synthetic data only	Original training data	Basic oversampling
Highest accuracy	58.57%	58.57%	54.29%	55.71%
F1 Score	0.556	0.560	0.474	0.515

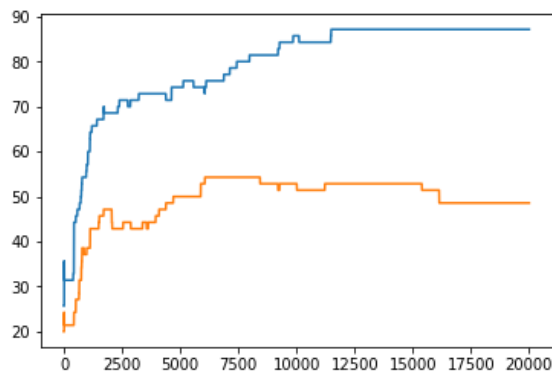
Figure 4 – Test set accuracy and F1 score on the petrographical dataset used by Scott (2019).

Simple neural network trained on original training data

Loss

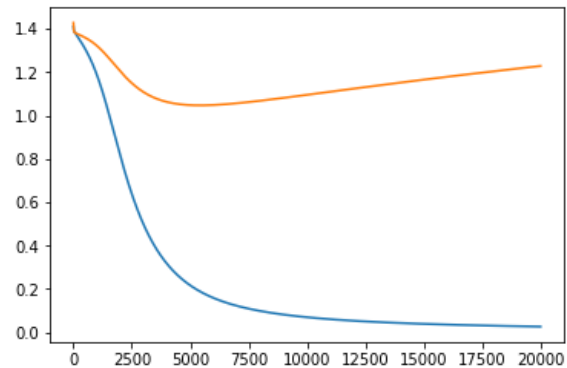


Accuracy



Simple neural network trained on synthetic training data produced by GAN-SMOTE

Loss



Accuracy

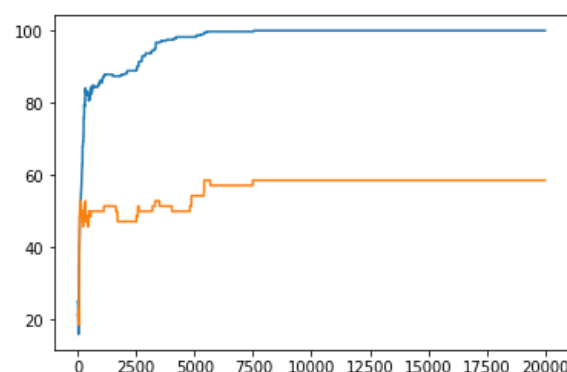


Figure 5 – (Left) Simple neural network trained on original training, training and testing loss and accuracy over 20,000 epochs; (Right) Simple neural network trained on synthetic data produced by GAN-SMOTE, training and testing loss and accuracy over 20,000 epochs. [Training data in blue, testing data in orange.]

These results show a substantial increase in training accuracy and F1 score for both tests involving GAN-SMOTE synthetic data. The small increase associated with basic oversampling also suggests that at least some of the improvement can be attributed to solving the class imbalance in the original training dataset, but the variance in the GAN-SMOTE synthetic data may also be playing a large role in the performance improvement.

The synthetic data only test demonstrated a remarkable level of training stability which is probably owing to the much larger number of datapoints. More stable training data allows neural network to achieve better results more consistently, with less hyperparameter tweaking. It seems that a basic network trained with synthetic data will overfit slower than a network trained with the original data. This can be seen in Figure 5.

4 Conclusion and Future Work

This paper proposed a novel method for dealing with class imbalance, utilising Generative Adversarial Networks to synthesise data for the purpose of oversampling minority classes in unbalanced datasets. GAN-SMOTE leverages a number of techniques to ensure that the GAN remains stable during training, and produces the correct amount of variance in the dataset to be useful for oversampling. The benchmark tests on the synthesised one-hot encoded data suggests that GAN-SMOTE can increase classifier performance, in terms of both raw accuracy and class-wise performance metrics.

The main benefit of this technique over other statistical techniques is the ability for the GAN to synthesise more realistic structured data than the traditional SMOTE technique can. For instance, this technique could be used to synthesise text or image data to boost domain specific datasets and will benefit from ongoing research into GAN generation of these datatypes.

There still remains a lot of work to discover the best practices when utilising this technique. GANs are difficult to train, and the rules of thumb discussed here may apply in different degrees to other types of data. Many changes needed to be made to the GAN-SMOTE architecture to account for the one-hot encoded dataset, so a robust understanding of the raw data being fed into the GAN will allow data scientists to use this technique successfully and repeatably.

Another area for investigation is the potential use of this technique to create synthetic data in scenarios with small datasets but no class imbalance. The improved training stability and reduced propensity to overfit can be a viable alternative to simply adding noise to training data because the variance produced by GAN-SMOTE is contextual. The technique appears to produce ‘acceptable’ variance while retaining key features, which is highly beneficial for datasets of high dimensionality such as this one.

Future work should focus on applying well-known deep learning techniques to the GAN-SMOTE architecture when dealing with different types of data. Advanced transfer learning techniques, such as universal language model fine-tuning for text classification (Howard & Ruder, 2018) could be applied to the discriminator and generator so that synthetic data benefits from the context of other similar problems. Transfer learning is a common tool to combat the problems associated with small datasets, and GAN-SMOTE could help to apply these benefits in a unique way.

5 References

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